Potentials & Equipotentials

Calculations, Equations, Product Modules, and Treatise Page Numbers

Full inventory: electric scalar potential V, equipotential surfaces, magnetic scalar potential, magnetic vector potential A, conduction/EMF potential language, product code paths, and OCR page IDs from the MAXWELL-MODERNIZED-PROGRAM product and page verifier catalog.

Page numbering: `v1-pNNN` = Volume I OCR/PDF page NNN; `v2-pNNN` = Volume II. These match `page_NNN.png` photos and the page verifier catalog (`volume_*_direct_result.json`).

Product root: `MAXWELL-MODERNIZED-PROGRAM/maxwell/`. Generated for human verification and documentation.

1. The big picture — types of potential

In Maxwell's *Treatise* (and in the product package) there are several different potentials. Equipotential means a surface of constant scalar electric potential (and the geometry of those surfaces).

Kind	Symbol	What it is	Where in Treatise
Electric (scalar) potential	V or psi	Work per unit charge; E = -grad V	Vol I, Part I
Equipotential surface	V = const	Surface of constant V; E perpendicular to it	Vol I (Art. 46 + forms of surfaces)
Magnetic scalar potential	Omega	Magnetostatics (current-free regions)	Vol II, Part III
Magnetic vector potential	A	B = curl A	Vol II, Arts. ~405-406+
Contact / voltaic / EMF potential	EMF, contact V	Chemistry/contact electricity, not Laplace theory	Vol I, Part II
Competing theories	Weber / Neumann	Historical alternatives	Late Vol II

2. Core electric potential equations

Main electrostatic potential calculations under `maxwell.core.potential`, `maxwell.core.field`, and supporting math. Units in product: **CGS-ESU** (statvolt, esu, cm).

Concept	Equation (CGS)	Articles	OCR pages	Product
Potential / EMF unit	units of potential	22-23	v1-p057, v1-p060	core.measurement.potential_unit
EMF as potential difference	work / unit charge along path	45	v1-p083	electromotive_force_potential
Equipotential surface	V=const; E perp surface	46	v1-p084	core.field.EquipotentialSurface
Line of force (related)	dx/Ex = dy/Ey = dz/Ez	47	v1-p086	core.field.LineOfForce
Line integral of intensity	integral E · dl	69	v1-p111	line_integral (+ JAX)
Definition of V	work bringing unit + charge from infinity	70	v1-p112	ElectricPotential, potential_difference
Point charge	V = q / r	70	v1-p112	ElectricPotential.from_point_charge
Field from potential	E = -grad V	71	v1-p113	field_from_potential (+ JAX)
Conductor equipotential	V constant on/in conductor	72	v1-p113	ElectricPotential.conductor_surface
Superposition	V = sum q_i/r_i	73	v1-p115	ElectricPotential.from_charges
Laplace (no charge)	nabla^2 V = 0	77	v1-p124	laplace_equation, solve_laplace
Poisson (with charge)	nabla^2 V = -4 pi rho	77	v1-p124	poisson_equation, solve_poisson
Boundary conditions	cont. V; jump in dV/dn	78	(no OCR marker)	boundary_condition_*
Mean-value of potential	sphere average	83	(no OCR marker)	potential_mean_value

Concept	Equation (CGS)	Articles	OCR pages	Product
Energy via potentials	system energy	85-88, 91-92	v1-p140+	system_energy, general_theorems.*
Potential of distribution	integral form	100-101	(no OCR marker for 100)	potential_from_charge_distribution
Uniqueness	solution unique under BCs	102	(no OCR marker)	uniqueness_theorem

Operators used with these: gradient, divergence, laplacian in `maxwell.math.vector_operators`; partials in `math.derivatives`; integrals in `math.calculus_calculator`.

3. Equipotentials — yes, that is the name

Maxwell's language: **equipotential surfaces** (and forms of those surfaces together with lines of force). An equipotential is a surface at every point of which the potential is the same; the electric field is everywhere perpendicular to an equipotential surface (Art. 46).

Topic	Articles	OCR pages	Product
Early viz / related phenomena cites	16-19 (vis module labeling)	v1-p049, 051, 052, 053	vis.equipotential.plot_*; examples/02_equipotential_surfaces.py
Definition of equipotential surface	46	v1-p084	EquipotentialSurface
Equilibrium points / saddle structure	112-115	v1-p204, 205, 207…	equilibrium_points, saddle_point_analysis
Generate equipotential surface V=const	117-119	v1-p212-215	equipotential_surface
Surface charge / curvature on equipotentials	119-121	v1-p214-216	surface_charge_density, surface_curvature
Field lines perp equipotentials	122-123	v1-p217-218	field_line_tracing
Isolated sphere / concentric / coaxial V	124, 126, 127	v1-p221, 225, 226	isolated_sphere, concentric_spheres, coaxial_cylinders
Confocal ellipsoid / hyperboloid equipotentials	147-156	v1-p267-280	confocal_ellipsoid_potential, confocal_hyperboloid, …

Example to run:

```
python examples/02_equipotential_surfaces.py
# -> examples/output/equipotential.png
```

4. Method of images (potential near conductors)

Case	Articles	Pages	Function
Plane	171-173	v1-p301, 303, 305	image_point_charge_plane
Sphere	174-176	v1-p307, 308, 311	image_point_charge_sphere
System analysis	181	v1-p314	image_system_analysis

5. Magnetic scalar potential and vector potential (Vol II)

Topic	Equation / idea	Articles	Pages	Product
Magnetic element potential	dOmega from magnet element	385-386	v2-p036, 037	physics.potentials.calc_element_potential, calc_finite_potential
Vector potential	B = curl A	405	v2-p056	calc_B_from_vector_potential, gauge, magnetization to A
Relate scalar / vector A	relation Omega to A	406	v2-p057	relate_scalar_vector_potential, vector_potential_uniform_field
Loop / shell	solid angle to A; jump of Omega	421-422	v2-p069, 070	vector_potential_closed_curve, magnetic_shell_potential_jump
Current-sheet / shell	shell potential, sheet A	648-651	v2-p313-314	calc_magnetic_shell_potential, calc_sheet_vector_potential

Topic	Equation / idea	Articles	Pages	Product
Cylinder A	A inside cylindrical conductor	686-687	v2-p345	calc_cylinder_vector_potential

Also: `maxwell/electromagnetism/potentials/` (surfaces, multivalued, mutual_energy, directrix) and `vis/helicoidal_potentials.py` for advanced geometry / visualization.

6. Potential in conduction / EMF (not Laplace)

Maxwell also uses potential language for contact electricity and current flow. These are not the same as solving Laplace/Poisson for electrostatic V, but they appear in the product under potential-related names.

Topic	Articles	Pages	Product
Contact / voltaic EMF	246-272 band	v1-p402-429	emf.*, emf_bodies.*, contact_potential
Point source / dipole in 3D conductor	293-294	v1-p449-450	point_source_potential, dipole_potential
Potential on resistive plane	307-309	v1-p464-467	potential_distribution_plane
Network voltages (Kirchhoff)	275-280	v1-p434-438	network_solver.*

7. Instruments that measure potential

Electrometers, attracted disk, torsion balance — Arts. **210-228**, pages **v1-p355** … **v1-p384** (`electrostatics.instruments.*`).

8. How this shows up in the calculator UI

Calculator mode	Potential-related content
Equation Calculator	Sets involving A, E from potentials, energy (catalog Sets A-H, etc.)
Derivatives	Partial V / gradient / Hessian (Laplace-related)
Integrals	Volume integrals for charge to potential / energy style calcs
Theorems	Divergence / Stokes / Green (potential theory backbone)

The Streamlit calculator does **not** yet expose every ElectricPotential / equipotential / image-charge API as a dedicated UI page — those are mainly Python API + examples + vis. Launch: `streamlit run maxwell/calculator_ui.py --server.port 8502`.

9. Page join gaps (OCR did not mark the article)

These have product code but no page_* in the OCR index right now: **78, 83, 85, 100, 102** (and possibly neighbors). You can still open nearby pages (e.g. 77 → **v1-p124**, 86 → **v1-p140**) in the page verifier; a class-C style fix would attach markers later.

10. High-value pages to open in the page verifier

Page ID	Why
v1-p049-053	Early equipotential / force-line material + vis cites
v1-p083-086	EMF potential, equipotential Art. 46, lines of force
v1-p111-115	Line integral, V definition, E = -grad V, conductors, superposition
v1-p124	Laplace / Poisson
v1-p140+	Energy
v1-p204-226	Equilibrium + equipotential surface generators
v1-p267-280	Confocal equipotentials
v1-p301-314	Method of images
v2-p036-037	Magnetic element potential
v2-p056-057	Vector potential A

Page ID	Why
v2-p069-070	Loop / magnetic shell potential

In the HTML page verifier: Jump = article number (e.g. 112) or page id (e.g. v1-p212), or function search for `field_from_potential` / `equipotential_surface`. Server: `python run_page_verifier.py` → http://127.0.0.1:8765/

11. Key product modules (file paths)

Module path	Role
maxwell/core/potential.py	ElectricPotential, Laplace, Poisson, BC, energy
maxwell/core/field.py	EquipotentialSurface, LineOfForce, field_from_potential, line_integral
maxwell/vis/equipotential.py	2D equipotential contour plots
maxwell/electrostatics/equilibrium_surfaces.py	equipotential_surface generator, field lines, spheres/cylinders
maxwell/electrostatics/confocal_surfaces.py	Confocal equipotentials / ellipsoidal Laplace
maxwell/electrostatics/electric_images.py	Method of images potentials
maxwell/electrostatics/general_theorems.py	Energy, Green reciprocity, uniqueness, potential from charge
maxwell/math/potential_theorems.py	Flux, Gauss surface, mean value, EMF defs
maxwell/calculus/vector_potential.py	Magnetic vector potential A
maxwell/calculus/cyclic.py	Loop A via solid angle; magnetic shell jump
maxwell/physics/potentials.py	Magnetic element / finite magnet potential
maxwell/electromagnetism/potentials/*	Advanced EM potential surfaces / multivalued / energy
maxwell/jax/…	JAX ports of potential/field helpers

12. Bottom line

- **Potentials** in this project = mainly scalar electric V, equipotentials V=const, then magnetic scalar Omega and vector A, plus conduction/EMF potential language.
- **Equipotentials** are first-class: Art. 46 (definition), Arts. 117+ (forms/surfaces), confocal 147+, plus vis/equipotential.
- **Page numbers** are from the OCR catalog join (article markers to volume PDF pages). A few articles still lack page markers (78, 83, 85, 100, 102).
- Treatise structure reminder: **2 volumes** (books), **4 scientific parts** (I-IV), continuous article numbers ~27-866 for the body of the work.

Document source: product code inventory + page_verifier citation/page index + Maxwell Treatise structure. Not a substitute for the original 1873/Third Edition text. Full response packaged as PDF per user request.